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French

Exercice1

On se propose de réaliser un programme permettant de manipuler des formes géométriques.

- 1) Définir une structure *Point* permettant de représenter un point du plan, par deux éléments réels x et y . Et ceux, pour contenir respectivement l'abscisse et l'ordonnée du point.
- 2) Écrire une fonction *DistancePoint*, permettant de calculer la distance entre deux points.
- 3) Définir une structure *Disque* permettant de représenter un disque, par un point et un rayon.
- 4) Ecrire une fonction *AireDisque*, permettant de calculer l'aire d'un disque.
- 5) Définir une structure *Triangle*, permettant de représenter un triangle par trois points $A(x_A, y_A)$, $B(x_B, y_B)$ et $C(x_C, y_C)$.
- 6) Ecrire la fonction *AireTriangle*, permettant de calculer l'aire d'un triangle défini par trois sommets (A, B, C) . On rappelle que $AireTriangle = 1/2 |(x_B - x_A)(y_C - y_A) - (x_C - x_A)(y_B - y_A)|$.
- 7) Définir une structure *Rectangle*, permettant de représenter un rectangle par deux points ; point supérieur gauche et point inférieur droit.
- 8) Ecrire la fonction *AireRectangle*, permettant de calculer l'aire d'un rectangle.

Ecrire un programme principal de test avec un menu de choix.

English

Exercise 1

We propose to create a program to manipulate geometric shapes.

- 1) Define a *Point* structure allowing to represent a point of the plane, by two real elements x and y . And those, to contain respectively the abscissa and the ordinate of the point.
- 2) Write a function *DistancePoint*, allowing to calculate the distance between two points.
- 3) Define a *Disk* structure allowing to represent a disk, by a point and a radius.
- 4) Write a function *AreaDisc*, allowing to calculate the area of a disc.
- 5) Define a *Triangle* structure, allowing to represent a triangle by three points $A(x_A, y_A)$, $B(x_B, y_B)$ and $C(x_C, y_C)$.
- 6) Write the function *AreaTriangle*, allowing to calculate the area of a triangle defined by three vertices (A, B, C) . We recall that $AireTriangle = 1/2 |(x_B - x_A)(y_C - y_A) - (x_C - x_A)(y_B - y_A)|$.
- 7) Define a *Rectangle* structure, allowing to represent a rectangle by two points; upper left point and lower right point.
- 8) Write the function *AreaRectangle*, allowing to calculate the area of a rectangle.

Write a main test program with a menu of choices.

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```

#include <stdio.h>
#include <stdlib.h>
#include <math.h>

typedef struct Point
{
    float x, y;
} POINT;

POINT A, B;

void get_points()
{
    printf("Enter Values of A(x, y):\n");
    printf("x = ");
    scanf("%f", &A.x);
    printf("y = ");
    scanf("%f", &A.y);
    printf("Enter Values of B(x, y):\n");
    printf("x = ");
    scanf("%f", &B.x);
    printf("y = ");
    scanf("%f", &B.y);
}

float distance(POINT A, POINT B)
{
    float distance;
    distance = sqrt(
        fabs(
            (B.x - A.x) * (B.x - A.x)) +
        fabs(
            (B.y - A.y) * (B.y - A.y)));
    return distance;
}

// fin_____\\Begin

typedef struct Cycle
{
    POINT original;
    float radius;
} CYRCLE;

CYRCLE c;

```

```

CYRCLE get_cyrcle()
{
    printf("Enter Values of A(x, y):\n");
    printf("x = ");
    scanf("%f", &c.original.x);
    printf("y = ");
    scanf("%f", &c.original.y);
    printf("Enter Values of B(x, y):\n");
    printf("x = ");
    scanf("%f", &c.original.x);
    printf("y = ");
    scanf("%f", &c.original.y);
    printf("Enter Values of Radius (r):\n");
    printf("r = ");
    scanf("%f", &c.radius);
}

float cyrcle_area(CYRCLE c)
{
    float area, PI = 3.14;
    area = PI * c.radius * c.radius;
    return area;
}

// fin_____\\Begin

typedef struct Triangle
{
    POINT A, B, C;
} TRIANGLE;

TRIANGLE t;

TRIANGLE get_triangle()
{
    printf("Enter Values of A(x, y):\n");
    printf("x = ");
    scanf("%f", &t.A.x);
    printf("y = ");
    scanf("%f", &t.A.y);
    printf("Enter Values of B(x, y):\n");
    printf("x = ");
    scanf("%f", &t.B.x);

```

```
printf("y = ");
scanf("%f", &t.B.y);
printf("Enter Values of C(x, y):\n");
printf("x = ");
scanf("%f", &t.C.x);
printf("y = ");
scanf("%f", &t.C.y);
```

```
}
```

```
float triangle_area(TRIANGLE t)
```

```
{
```

```
    float area, part1, part2;
    part1 = (t.B.x - t.A.x) * (t.C.y - t.A.y);
    part2 = (t.C.x - t.A.x) * (t.B.y - t.A.y);
    area = fabs(part1 - part2) / 2;
    return area;
```

```
}
```

```
// fin_____\\Begin
```

```
typedef struct Rectangle
```

```
{
```

```
    POINT topLeft, buttonRight;
```

```
} RECTANGLE;
```

```
RECTANGLE r;
```

```
RECTANGLE get_rectangle()
```

```
{
```

```
    printf("Enter Values of top left A(x, y):\n");
    printf("x = ");
    scanf("%f", &r.topLeft.x);
    printf("y = ");
    scanf("%f", &r.topLeft.y);
    printf("Enter Values of button right B(x, y):\n");
    printf("x = ");
    scanf("%f", &r.buttonRight.x);
    printf("y = ");
    scanf("%f", &r.buttonRight.y);
```

```
}
```

```
float rectangle_area(RECTANGLE r)
```

```
{
```

```
    float area;
```

```

    area = fabs(
        (r.buttonRight.x - r.topLeft.x) *
        (r.topLeft.y - r.buttonRight.y));
    return area;
}

int main()
{
    int choice;
    printf("\n\n");
    do
    {
        printf("\nEnter choice: \n");
        printf("1 -----> Points,\n");
        printf("2 -----> Cyrcle,\n");
        printf("3 -----> Triangle,\n");
        printf("4 -----> Rectangle,\n");
        printf("5 -----> Exit,\n");
        printf("\nChoice: ");
        scanf("%d", &choice);
        switch (choice)
        {
            case 1:
                get_points();
                printf("\nDistance = %.2f\n", distance(A, B));
                break;
            case 2:
                get_cyrcle();
                printf("\nArea of the Cyrcle = %.2f\n", cyrcle_area(c));
                break;
            case 3:
                get_triangle();
                printf("\nArea of the Triangle = %.2f\n", triangle_area(t));
                break;
            case 4:
                get_rectangle();
                printf("\nArea of the Rectangle = %.2f\n", rectangle_area(r));
                break;
            default:

```

```
        break;
    }
} while (choice < 5);
return 0;
}
```

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